

YOUNG'S MODULUS OF A COPPER WIRE (6)

Aim

To determine the Young's Modulus of a Copper wire

Materials and Apparatus

Copper wire, meter ruler, cathetometer, micrometer screw gauge, weights, stands, loop

Theory

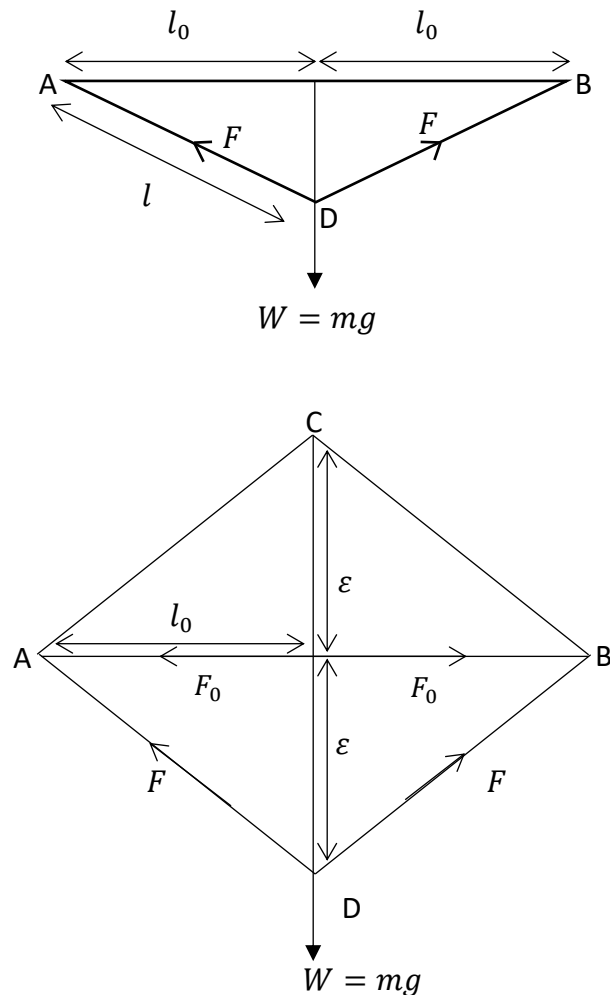


Fig. 1: The figures of the force triangle and force parallelogram

Where; ϵ – Displacement of the wire, F_0 , F – stretching force of the wire before and after loading the weights, $2l_0$, $2l$ – length of the wire before and after loading the weights, m – mass, g – acceleration due to gravity

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Let ADC be a triangle of force.
From the triangle of forces law,

$$\frac{F}{l} = \frac{W}{2\varepsilon} \quad (1)$$

$$F = \frac{Wl}{2\varepsilon}$$

Also,

$$l = (l_0^2 + \varepsilon^2)^{1/2}$$

From Binomial expansion,

$$l = l_0 \left[1 + \frac{\varepsilon^2}{2l_0^2} \right]$$

$$l - l_0 = \frac{\varepsilon^2}{2l_0} \quad (2)$$

But Young's Modulus,

$$E = \left\{ \frac{Wl_0/2\varepsilon - F_0}{A} \right\} / \left\{ \frac{l - l_0}{l_0} \right\} \quad (3)$$

where, A - Cross-sectional area of the wire

E - Young's Modulus

Equations (2) & (3);

$$E = \frac{[Wl_0/2\varepsilon - F_0] 2l_0^2}{A\varepsilon^2}$$

$$EA\varepsilon^2 = \frac{Wl_0^3}{\varepsilon} - 2F_0l_0^2$$

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$$\frac{W}{\varepsilon} = \left[\frac{EA}{l_0^3} \right] \varepsilon^2 + \frac{2F_0}{l_0}$$

This yields a straight line of the form $y = mx + c$

From the graph of W/ε vs. ε^2

The gradient

$$m = \frac{EA}{l_0^3}$$

$$E = \frac{ml_0^3}{A}$$

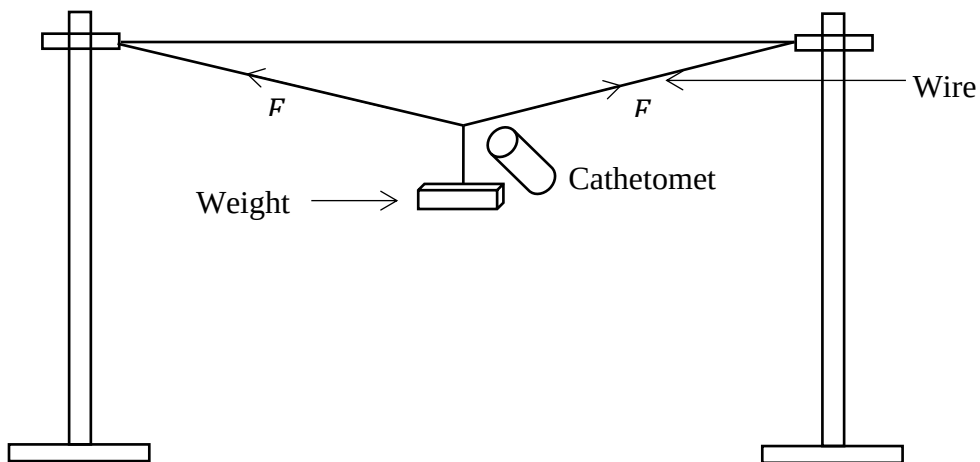


Fig. 2: The schematic diagram of the setup

Procedure:

1. Fix the Copper wire horizontally and tightly as shown in Fig 2.
2. Attach a loop to the midpoint of the wire and take the cathetometer reading of the midpoint.
3. Add weights to the loop starting from 50 g up to 100 g and measure the corresponding displacement of the Copper wire using the cathetometer.
4. Repeat the procedure while unloading the weights.
5. Measure the length and the diameter of the wire.

(Use safety goggles when doing the experiment)